

Dataset Metadata

What each file in the starter kit looks like, what it contains, where it came from, and how it is meant to be used.

Covers Every file in data/team_a_nairobi and data/team_b_nzoia | **Coordinates** Latitude and longitude, WGS84 (EPSG:4326)
| **Currency** Kenyan shillings (KES)

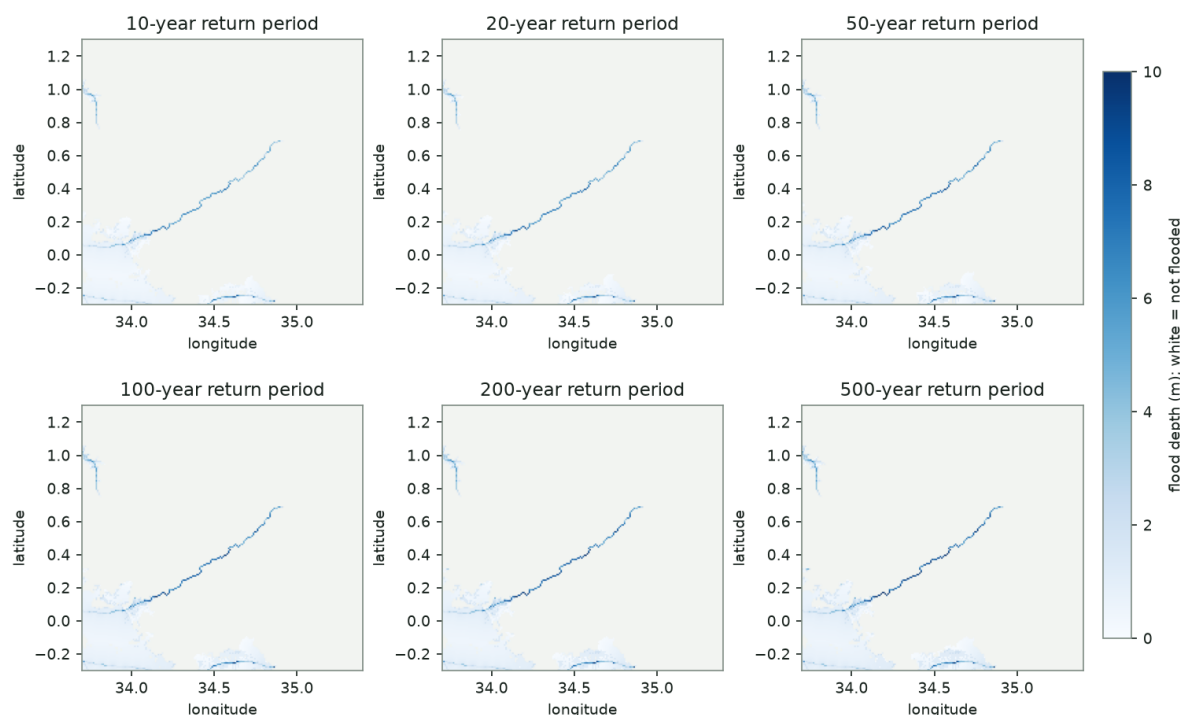
1. Read this first

Each dataset below is described the same way: a picture or sample rows, a plain explanation of what you are looking at, how to use it, and a table of technical detail. Two of the datasets are measured or modelled by outside organisations. The rest were built for this hackathon, and each of those says so in its own data as well as here.

Team	File	Real or synthetic	Holds
Team B	nzoia_rp10y, rp20y, rp50y, rp100y, rp200y, rp500y (.tif)	Real	Flood depth maps
Team B	exposure_nzoia_synthetic.csv	Synthetic	500 buildings
Team A	nairobi_pluvial_proxy_common, occasional, moderate, severe, extreme (.tif)	Built from real terrain and river data; the score is a constructed proxy	Flood susceptibility maps
Team A	nairobi_hotspots_geocoded.csv	Real names; approximate coordinates	24 named flood areas
Team A	exposure_nairobi_synthetic.csv	Synthetic	600 buildings
Team A	exposure_nairobi_with_hazard.csv	Synthetic buildings, hazard values read from the maps	Same 600 buildings plus 5 hazard columns

2. Team B: Nzoia Basin

Flood depth maps (six GeoTIFF files)



The six Nzoia files drawn on one depth scale. Each panel is one return period. Dark blue is deep water; the pale grey is dry ground.

WHAT YOU ARE LOOKING AT

Each file is a grid laid over the lower Nzoia basin. Every cell covers about 900 m by 900 m of ground, and its value is the depth of flood water in metres for one scenario. The scenario is named in the filename: rp10y is a flood expected about once in 10 years, rp500y about once in 500 years.

The pattern is narrow. Water shows up along the Nzoia River, around its mouth at Lake Victoria in the south-west, and along a small channel in the south and in the north-west corner. Only about 9% of cells are flooded in any of the six maps, and the flooded area grows very little between the 10-year and 500-year maps: from 8.7% to 9.8% of cells, with median depth rising from 0.49 m to 0.74 m. Expect a loss curve that rises sharply once and then climbs slowly.

HOW TO USE IT

- Take a building's longitude and latitude and find the cell that contains it in each of the six files.
- Read the cell value from each file. You now have six depths for that building, one per return period.
- Dry cells hold a no-data value, a very large negative number (about -3.4×10^{38}). They are never stored as 0. Treat any value that is not a positive number as dry, and do not average or sum them.
- Pass each depth through your vulnerability function to get a damage ratio.

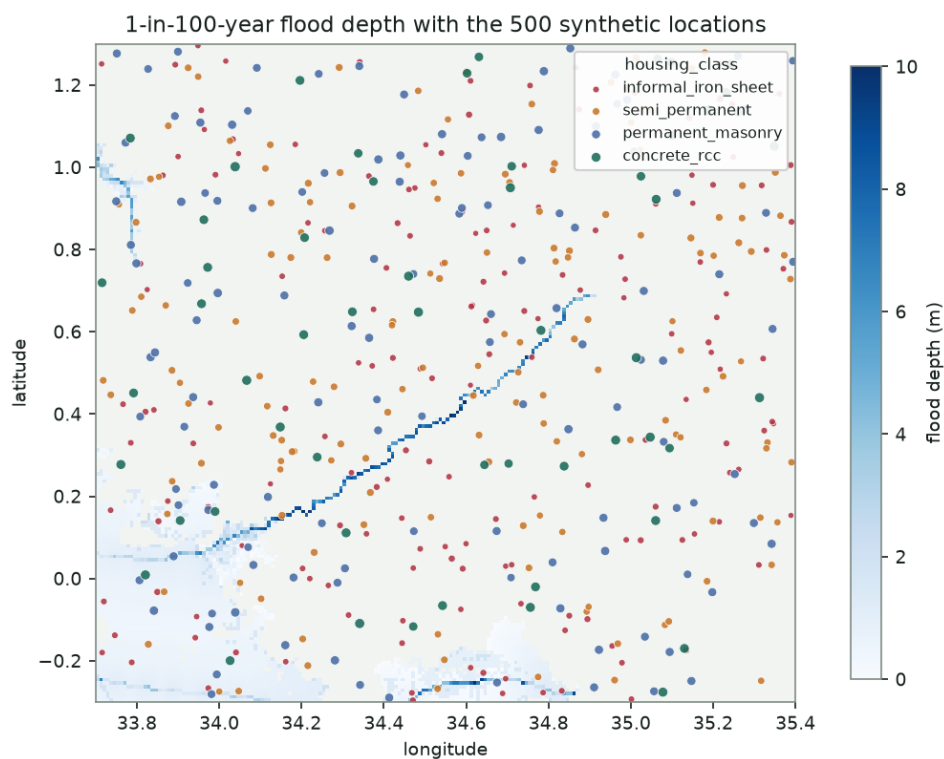
Field	Value
Provider	European Commission Joint Research Centre (JRC)
Product	Global river flood hazard maps, file set floodMapGL_rp10y to rp500y. The files in the downloaded archives are dated 22 November 2016

Licence	Free for public use. See the JRC data catalogue page for the exact terms
Format	GeoTIFF, one band, 32-bit float
Cell size	0.00833° (30 arc-seconds), about 925 m
Size of this clip	204 columns × 192 rows, longitude 33.70 to 35.40, latitude -0.30 to 1.30
Cell values	Flood depth in metres. Dry cells = no-data value
Flooded cells	3,721 of 39,168 in the 100-year map (9.5%)
Deepest cell	13.1 m in the 100-year map
Limits	Coarse for a single building. A 925 m cell cannot tell a house on a rise from one in a hollow, and small drains are not represented

Synthetic portfolio: exposure_nzoia_synthetic.csv

loc_id	lat	lon	housing_class	floor_area_m2	cost_per_m2_kes	tiv_kes
NZA-0000	0.7566	33.972	concrete_rcc	489	56,000	27,385,000
NZA-0001	0.03	34.6961	informal_iron_sheet	8	9,600	75,000
NZA-0002	1.0777	35.1005	concrete_rcc	385	72,000	27,720,000
NZA-0003	0.593	34.2053	concrete_rcc	519	51,200	26,575,000
NZA-0004	0.8532	34.8643	permanent_masonry	97	52,500	5,090,000
NZA-0005	0.51	34.4886	informal_iron_sheet	13	7,700	100,000

First six of 500 rows. The file also has 'synthetic' and 'source' columns that state, on every row, that the building was generated and is not a real property. *tiv_kes* is *floor_area_m2* multiplied by *cost_per_m2_kes*, rounded to the nearest 5,000.



The 500 buildings placed on the 100-year flood depth map. Dot colour is housing class; dot size is insured value.

WHAT YOU ARE LOOKING AT

Five hundred invented buildings scattered at random across the same area as the flood maps. Nothing about them was measured. Locations are random, housing classes are assigned in fixed proportions, and each value is a floor area multiplied by a building cost per square metre, both chosen to suit the housing class (see 'How the insured values were set' below). Because placement is random and flooding is confined to a narrow corridor, only 46 to 48 of the 500 buildings sit in a flooded cell, depending on the map.

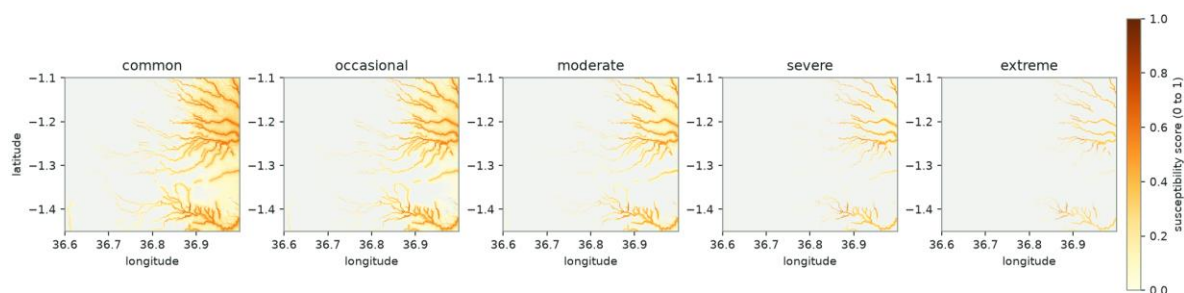
HOW TO USE IT

- Join it to the flood maps by location, as described above.
- Use housing_class to choose which damage curve applies to each building, and tiv_kes as the value that the damage ratio is multiplied by.
- Keep the 'synthetic' column through any filtering or merging, so the label survives to your results.
- You may regenerate or extend the portfolio. If you do, say so, and do not describe any version of it as real client data.

Field	Value
Rows	500
Columns	loc_id, lat, lon, housing_class, floor_area_m2, cost_per_m2_kes, tiv_kes, synthetic, source
loc_id	Text identifier, NZA-0000 to NZA-0499
housing_class	informal_iron_sheet (168), semi_permanent (154), permanent_masonry (124), concrete_rcc (54)
tiv_kes	Total insured value in KES, the cost to rebuild the structure. Minimum 45,000; median 490,000; maximum 72,435,000; total 2,273,710,000
Real or synthetic	Synthetic. Generated for this hackathon with a fixed random seed

3. Team A: Nairobi

Flood susceptibility maps (five GeoTIFF files)



The five Nairobi tiers side by side. Darker orange means a higher susceptibility score. Each tier keeps fewer cells than the one before it.

WHAT YOU ARE LOOKING AT

No measured flood depth exists for Nairobi, so these maps score how likely each 30 m cell is to collect water, on a scale from 0 to 1. A score is not a depth. Do not describe it as one, and do not convert it to metres without stating the assumption you used.

The score blends four real signals: how low the cell sits compared with all of Nairobi (45%), whether it is a local dip in the ground (20%), how flat the ground is (15%), and how close it is to a mapped river or stream (20%). The brown channels in the pictures are the river valleys, which is where the score is highest.

The five files are cuts through that one score. 'common' keeps the highest-scoring 40% of cells, 'occasional' 30%, 'moderate' 20%, 'severe' 10% and 'extreme' 5%, each rescaled to run from 0 to 1. That makes the tiers a ranking of how extreme the cell is. They do not carry return periods of their own. The names describe how extreme a cell is, not how often it floods. A rarer flood reaches more places, so the widest map ('common') stands for the rarest event and the narrowest ('extreme') for the most frequent. The reference dashboard assumes 10 years for 'extreme', 25 for 'severe', 50 for 'moderate', 100 for 'occasional' and 250 for 'common'. That is an assumption made so the loss curve can be drawn. State your own if it differs, and check that loss rises as the event gets rarer.

HOW TO USE IT

- Look up each building's cell in each of the five files, or skip this step with `exposure_nairobi_with_hazard.csv` (below), which already holds the five values per building.
- Dry cells are exactly 0 in these files. There is no separate no-data value.
- Decide how a 0 to 1 score becomes damage. The reference dashboard feeds the score straight into a saturating curve. You could instead map the score to an assumed depth, or work in damage bands. Either is fine if you say which you chose.
- Decide which return period each tier stands for, and say so. Remember that the widest footprint is the rarest event.

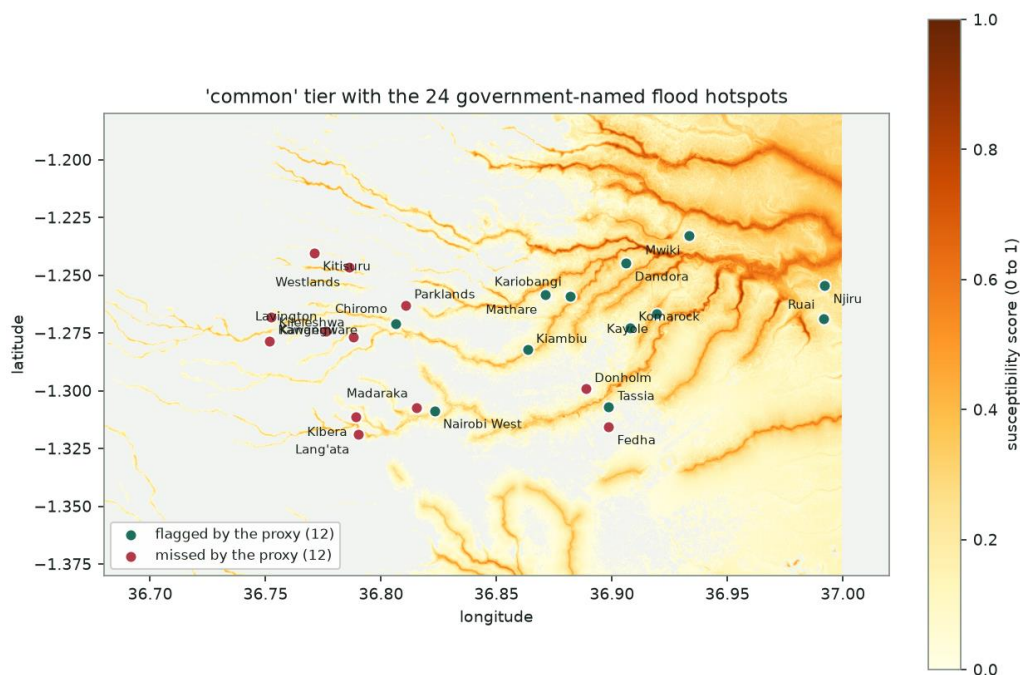
Field	Value
Built by	This project, from open data. No public pluvial hazard dataset exists for Nairobi
Inputs	Copernicus GLO-30 elevation model; river and stream lines from OpenStreetMap
Format	GeoTIFF, one band, 32-bit float. Values 0 to 1; dry = 0
Cell size	0.000278° (1 arc-second), about 31 m
Size	1,439 columns × 1,260 rows, longitude 36.60 to 37.00, latitude -1.45 to -1.10
Why it stops at 37.00	The elevation tile it was cut from ends at longitude 37.0, so the strip from 37.00 to 37.05 has no data
Cells above zero	common 40%, occasional 30%, moderate 20%, severe 10%, extreme 5%
Checked against	24 government-named flood areas in Nairobi County. The 'common' tier flags 12 of the 24
Limits	Terrain and rivers only. It knows nothing about rainfall, blocked drains, or the state of the stormwater network, which cause much of Nairobi's flooding

Named flood hotspots: `nairobi_hotspots_geocoded.csv`

name	lat	lon	name	lat	lon
Kiambu	-1.2823	36.8634	Nairobi West	-1.3087	36.8231
Dandora	-1.2449	36.9061	Lang'ata	-1.319	36.79
Kariobangi	-1.2591	36.8819	Kawangware	-1.2785	36.7516
Kayole	-1.2667	36.9194	Kangemi	-1.268	36.7523
Komarock	-1.2727	36.9077	Lavington	-1.2741	36.7762
Njiru	-1.2546	36.9921	Westlands	-1.2465	36.7861
Ruai	-1.269	36.9915	Parklands	-1.2631	36.8106

Mwiki	-1.2328	36.9333	Kitisuru	-1.2404	36.771
Donholm	-1.2992	36.8887	Kileleshwa	-1.2767	36.7879
Tassia	-1.307	36.8986	Chiromo	-1.2708	36.8064
Fedha	-1.3157	36.8985	Mathare	-1.2584	36.8713
Madaraka	-1.3074	36.8153	Kibera	-1.3113	36.789

All 24 rows, shown in two columns to save space.



The hotspots on the 'common' tier. Green dots fall on a cell the proxy scores above zero; red dots do not.

WHAT YOU ARE LOOKING AT

The names are real: they come from Nairobi County's own list of flood-prone neighbourhoods. The coordinates are not official. Each name was looked up automatically in OpenStreetMap's place search and is the middle of the area, not the exact spot that floods. The county named 37 areas; this file holds the 24 that could be located.

The map is the proxy's report card. The twelve green hotspots (Kayole, Njiru, Kiambiu, Mwiki, Mathare, Dandora, Tassia, Kariobangi, Nairobi West, Ruai, Komarock and Chiromo) lie in the river valleys that the proxy picks out. The twelve red ones, mostly in the west of the city, flood for reasons the proxy cannot see, such as drainage capacity.

HOW TO USE IT

- Use it to test any hazard layer you build: a sensible layer should score these places higher than the city average.
- Do not treat it as exposure. These are named areas, not buildings.
- Report your own hit rate against it in the same plain terms used here.

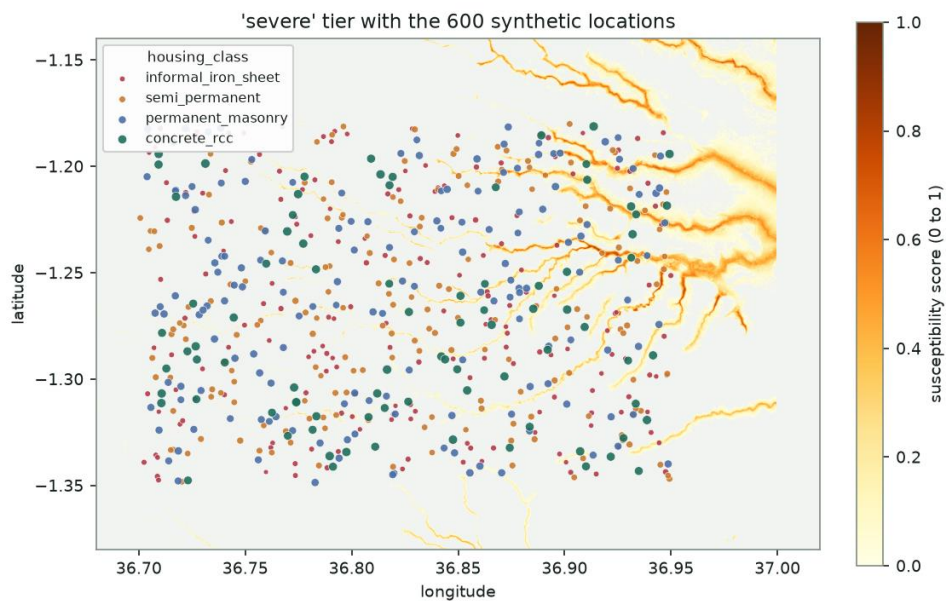
Field	Value
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Rows / columns	24 rows; name, lat, lon
Source of names	Nairobi County flood-hotspot mapping, reported March 2026
Source of coordinates	OpenStreetMap Nominatim place search. Approximate
Real or synthetic	Real names, approximate coordinates

Synthetic portfolio: exposure_nairobi_with_hazard.csv

loc_id	lat	lon	housing_class	floor_area_m2	cost_per_m2_kes	tiv_kes	common	occasional	moderate	severe	extreme
NBO-0000	-1.3149	36.9359	semi_permanent	38	13,600	515,000	0.023	0.0	0.0	0.0	0.0
NBO-0001	-1.22	36.7899	informal_iron_sheet	8	7,800	60,000	0.0	0.0	0.0	0.0	0.0
NBO-0002	-1.2576	36.8962	semi_permanent	33	10,000	330,000	0.458	0.407	0.345	0.247	0.135
NBO-0003	-1.1941	36.8478	semi_permanent	58	13,700	795,000	0.114	0.03	0.0	0.0	0.0
NBO-0004	-1.2632	36.7736	semi_permanent	78	15,900	1,240,000	0.141	0.06	0.0	0.0	0.0
NBO-0005	-1.2023	36.9307	informal_iron_sheet	16	9,800	155,000	0.67	0.639	0.601	0.541	0.472

First six of 600 rows. The last five columns are hazard_score_common to hazard_score_extreme, shortened here to fit the page. A score of 0 means the building sits on a cell the proxy does not flag at that tier.



The 600 buildings on the 'severe' tier. Dot colour is housing class; dot size is insured value.

WHAT YOU ARE LOOKING AT

Six hundred invented buildings spread at random across Nairobi, each with the five hazard scores already read from the maps. Placement is random, so the portfolio is not concentrated where flooding is worst. 259

buildings have a score above zero in the 'common' tier, 174 in 'occasional', 110 in 'moderate', 51 in 'severe' and 32 in 'extreme'.

HOW TO USE IT

- Start here if you want to skip raster handling: each row already has the hazard value for every tier.
- Use `housing_class` to choose the damage curve and `tiv_kes` as the value at risk.
- Keep the 'synthetic' column. The same caution applies as for Nzoia: never present this as a real portfolio.
- `exposure_nairobi_synthetic.csv` is the same 600 buildings without the five hazard columns. Use it only if you are building your own hazard layer and need to look the values up again.

Field	Value
Rows	600 in each of the two Nairobi exposure files
Columns (with_hazard)	loc_id, lat, lon, housing_class, floor_area_m2, cost_per_m2_kes, tiv_kes, synthetic, source, hazard_score_common, hazard_score_occasional, hazard_score_moderate, hazard_score_severe, hazard_score_extreme
loc_id	Text identifier, NBO-0000 to NBO-0599
housing_class	semi_permanent (181), informal_iron_sheet (179), permanent_masonry (156), concrete_rcc (84)
tiv_kes	Total insured value in KES, the cost to rebuild the structure. Minimum 40,000; median 520,000; maximum 159,200,000; total 6,363,470,000
Real or synthetic	Synthetic buildings. The hazard columns are read from the proxy maps, so they carry the proxy's limits

4. Source data behind the Nairobi maps

Dataset	Detail
Copernicus GLO-30 elevation model	Global 30 m height model from the European Space Agency and the Copernicus programme. Taken from the public AWS copy, tile Copernicus_DSM_COG_10_S02_00_E036_00_DEM, which covers Nairobi. Free to use.
OpenStreetMap rivers and streams	Lines tagged 'river' or 'stream' in OpenStreetMap, fetched through the public Overpass service for the Nairobi area. Licensed under the Open Database Licence; anything you republish must credit OpenStreetMap contributors.

5. How the insured values were set

Each building's value is its floor area multiplied by a rebuilding cost per square metre. Both inputs depend on the housing class, and both are shown in the file (`floor_area_m2` and `cost_per_m2_kes`) so any value can be checked by hand. The values cover the structure only, not its contents or any land.

Housing class	Floor area	Cost per m ²	Published figure it was set against
informal_iron_sheet	8 to 28 m ² , about 14 typical	KES 5,000 to 10,000	A basic two-room unit is

			reported at about KES 80,000
semi_permanent	20 to 80 m ² , about 38 typical	KES 8,000 to 16,000	Semi-permanent houses are reported at KES 200,000 to 500,000
permanent_masonry	45 to 300 m ² , about 100 typical	KES 35,000 to 65,000	Formal construction costs KES 35,000 to 65,000 per m ² for basic to mid-range finishes; KES 54,730 for a standard Nairobi bungalow
concrete_rcc	Nairobi 250 to 4,000 m ² ; Nzoia 150 to 1,800 m ²	KES 50,000 to 85,000	Blocks, shops and institutions, so far larger than a house. Rural buildings are smaller

The cost figures come from published 2025 Kenyan construction guides (for example Integrum Construction's 'Construction Costs in Kenya'). The floor areas are invented but typical for each class. These are illustrative replacement costs, not valuations of any real building, and a real portfolio would carry the insured sums on the policy schedule instead.

6. Conventions

- Coordinates are decimal degrees, latitude and longitude, WGS84 (EPSG:4326), in both the rasters and the CSVs.
- Money is in Kenyan shillings.
- A synthetic file carries a 'synthetic' column set to True.
- Nzoia dry cells are no-data; Nairobi dry cells are 0.